

2013 HKDSE Physics

Paper 1A Multiple Choice Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Question 1

Which of the following statements about boiling and evaporation of a liquid is/are correct?

- (1) A liquid absorbs energy when it boils but does not absorb energy when it evaporates.
- (2) Boiling occurs at a definite temperature while evaporation takes place above room temperature.
- (3) Boiling occurs throughout the liquid while evaporation only takes place at the liquid's surface.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: B

Both processes require latent heat, so (1) is false. Evaporation can occur below room temperature, so (2) is false; at a fixed external pressure boiling occurs at the boiling point.

Vapour bubbles form throughout a boiling liquid, whereas evaporation is escape from the surface. Only (3) is correct: B.

Question 2

In an experiment to measure the specific latent heat of vaporization of water, a beaker of water is boiled off using an electric heater. Which source of error would lead to an experimental result smaller than the standard value?

A. Energy is lost to the surroundings.

B. Water splashes out of the beaker.

C. Steam condenses on the cooler part of the heater and drops back to the beaker.

D. The heater is not completely immersed in water.

Official answer: B

The calculation is $l = Pt/\Delta m$, treating the measured mass loss as evaporated water. Splashing makes Δm too large without requiring vaporization energy, so the calculated l is too small: B.

Heat loss and incomplete immersion waste input energy, tending to overestimate l . Returning condensate reduces measured mass loss, also increasing $Pt/\Delta m$.

Question 3

* In which situations would the r.m.s. speed of the molecules of a fixed mass of an ideal gas increase?

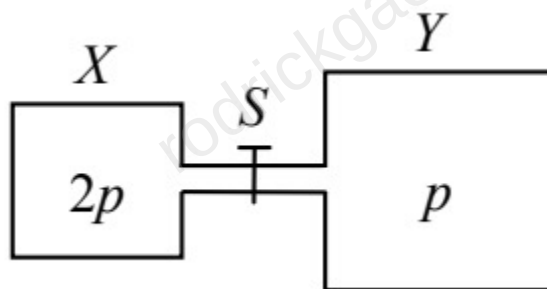
- (1) The gas is heated under constant volume.
- (2) The gas expands under constant pressure.
- (3) The gas is compressed under constant temperature.

- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

Official answer: C

For the same gas, $v_{\text{rms}} = \sqrt{(3RT/M)}$, so only a rise in absolute temperature increases r.m.s. speed.

(1) Heating at constant volume raises T . (2) At constant pressure and fixed mass, V/T is constant, so expansion raises T . (3) Isothermal compression leaves T and r.m.s. speed unchanged. Choose C.

Question 4

* Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube. Initially tap S is closed; the same ideal gas at the same temperature is contained in X and Y at pressures $2p$ and p respectively. S is opened and equilibrium is reached at unchanged temperature. Which statement is INCORRECT?

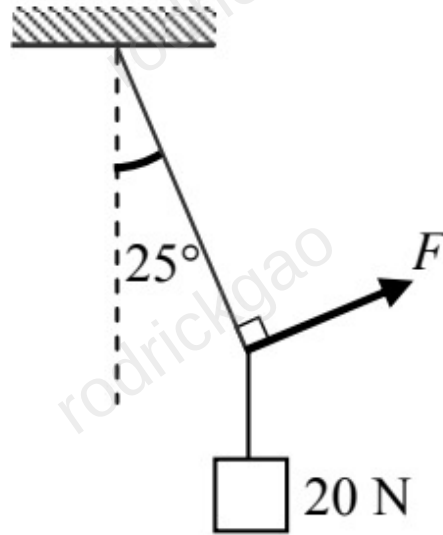
- A. Before S is opened, both vessels contain the same number of gas molecules.
 B. Before S is opened, the average kinetic energy of gas molecules in both vessels is the same.
 C. When S is opened, a net flow of gas from X to Y occurs.
 D. At equilibrium the gas pressure becomes $(3/2)p$.

Official answer: D

Initially $N_X kT = (2p)V$ and $N_Y kT = p(2V)$, so the numbers are equal. Equal temperature also means equal average molecular kinetic energy. The initial pressure difference drives a net flow from X to Y.

Conserve total molecules at fixed T : $p_f(3V) = 2pV + p(2V) = 4pV$, giving $p_f = 4p/3$, not $3p/2$. The incorrect statement is D; an unweighted mean of pressures is invalid for unequal volumes.

Question 5



A block of weight 20 N is suspended by a light string from the ceiling. A force F displaces the block to one side with the string making 25° with the vertical, as shown. Find F .

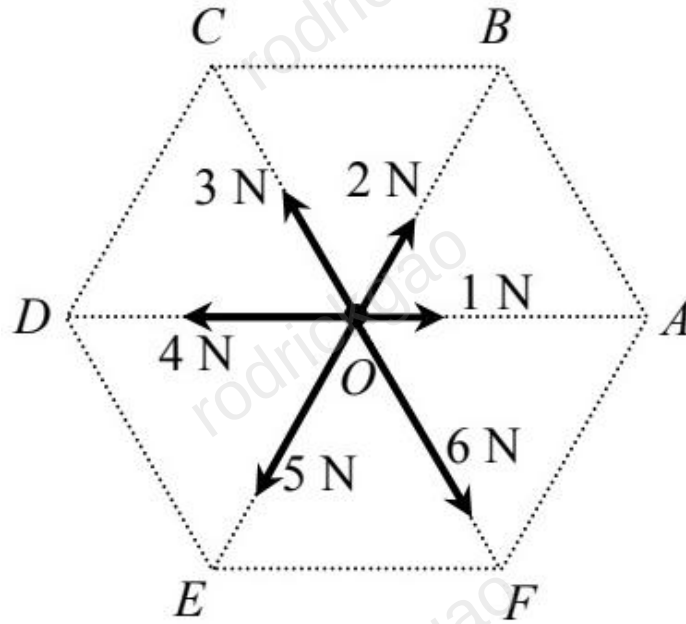
- A. 8.5 N
- B. 9.3 N
- C. 18.1 N
- D. 47.3 N

Official answer: A

The right-angle mark shows F perpendicular to the sloping string, not horizontal. Resolve forces perpendicular to the string: tension has no component, while the weight contributes $20 \sin 25^\circ$ opposite to F .

Equilibrium gives $F = 20 \sin 25^\circ = 8.45 \text{ N} \approx 8.5 \text{ N}$: A. Using $20 \tan 25^\circ$ would describe a horizontal applied force and does not fit the diagram.

Question 6



O is the centre of a regular hexagon. A particle at O is subject to the six forces shown. The resultant force on the particle is

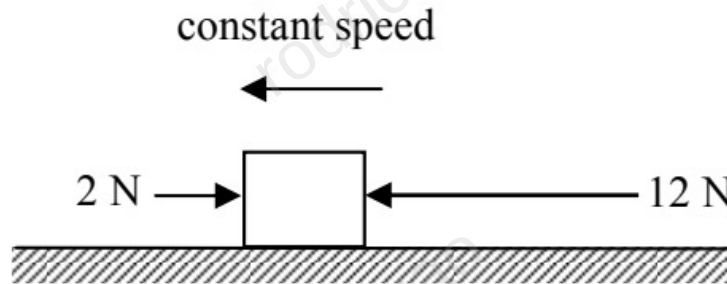
- A. 9 N along OE.
- B. 8 N along OE.
- C. 8 N along OF.
- D. 6 N along OE.

Official answer: D

First subtract opposite forces: $4 - 1 = 3$ N along OD, $5 - 2 = 3$ N along OE, and $6 - 3 = 3$ N along OF.

OD and OF are separated by 120° . Their equal 3 N vectors give $2(3)\cos 60^\circ = 3$ N along the bisector OE. Add the remaining 3 N along OE to obtain 6 N along OE: D.

Question 7



A block on a rough horizontal surface moves left with constant speed under the horizontal forces 2 N and 12 N shown. If the 12 N force is suddenly removed, what is the net force on the block at that instant?

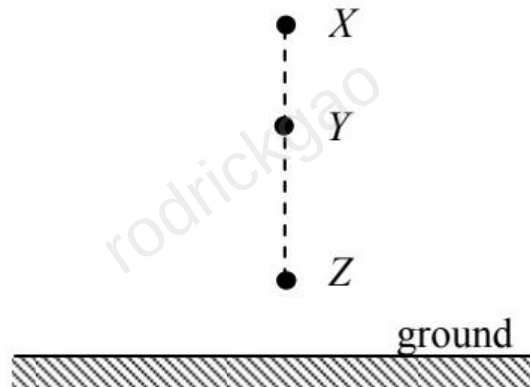
- A. 12 N
- B. 10 N
- C. 8 N
- D. 2 N

Official answer: A

Initially the net force is zero: friction is $12 - 2 = 10$ N to the right, opposing the leftward sliding.

Immediately after removing 12 N, the block still slides left and friction is still 10 N right. The remaining 2 N is also rightward, so the resultant is 12 N to the right: A. Friction does not disappear when an applied force is removed.

Question 8



A particle is released from rest at X. It takes time t_1 to fall from X to Y and time t_2 from Y to Z. If $XY : YZ = 9 : 16$, find $t_1 : t_2$. Neglect air resistance.

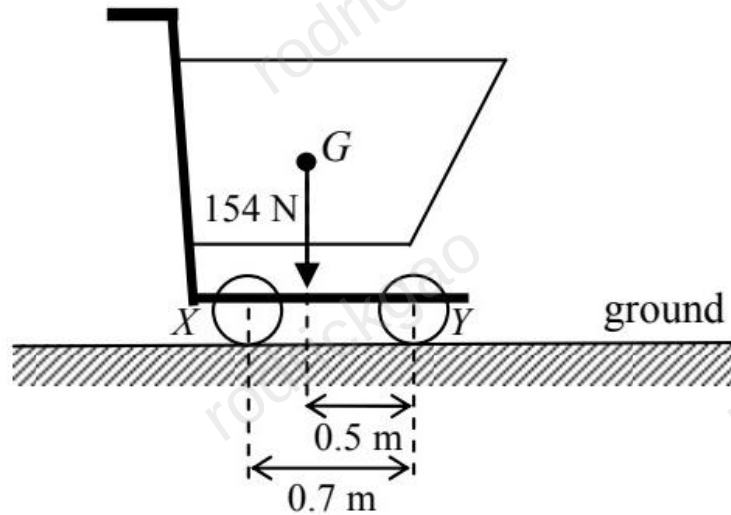
- A. 2 : 3
- B. 3 : 4
- C. 4 : 3
- D. 3 : 2

Official answer: D

Free fall from rest gives distance proportional to total elapsed time squared. $XY : XZ = 9 : 25$, so $t_1 : (t_1 + t_2) = 3 : 5$.

Thus $t_1 : t_2 = 3 : (5 - 3) = 3 : 2$, D. Do not apply $s = \frac{1}{2}gt^2$ separately to YZ: the particle is not at rest at Y.

Question 9



A supermarket trolley rests on the ground. Cylindrical wheels X and Y are 0.7 m apart. Its total loaded weight is 154 N and its centre of gravity G is a horizontal distance 0.5 m from Y. What is the ground reaction on wheel X?

A. 44 N

B. 62 N

C. 92 N

D. 110 N

Official answer: D

Take moments about Y to eliminate its unknown reaction. Rotational equilibrium gives $R_X(0.7) = 154(0.5)$, hence $R_X = 110$ N: D.

Check vertical equilibrium: $R_Y = 154 - 110 = 44$ N. The centre of gravity is closer to X, so X should carry the larger load.

Question 10

Two identical spheres move in opposite directions with speeds u and v ($u > v$) and make a head-on collision. Which diagrams show possible situations after collision?



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only**

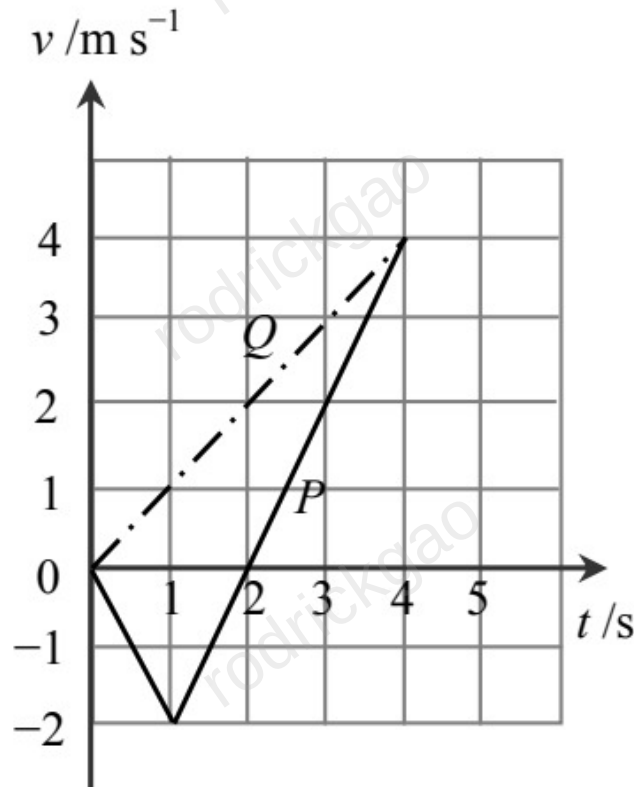
Official answer: D

Take right as positive. Initial momentum is $m(u-v) > 0$. In (1), final momentum is $-mu$, so it cannot be conserved. In (2), the spheres exchange velocities, conserving both momentum and kinetic energy: possible.

In (3), momentum conservation requires $mu - mv = mv$, or $u = 2v$, which is allowed by $u > v$. Kinetic energy decreases from $\frac{1}{2}m(4v^2 + v^2)$ to $\frac{1}{2}mv^2$, so an inelastic collision is possible. Therefore (2) and (3): D. The word possible does not mean for every u and v .

Question 11

Particles P and Q start at the same position and travel along the same straight line. Their velocity-time graph is shown. Which descriptions are correct?



- (1) At $t = 1$ s, P changes its direction of motion.
- (2) At $t = 2$ s, the separation between P and Q is 4 m.
- (3) At $t = 4$ s, P and Q meet.

A. (1) only

B. (2) only

C. (1) and (3) only

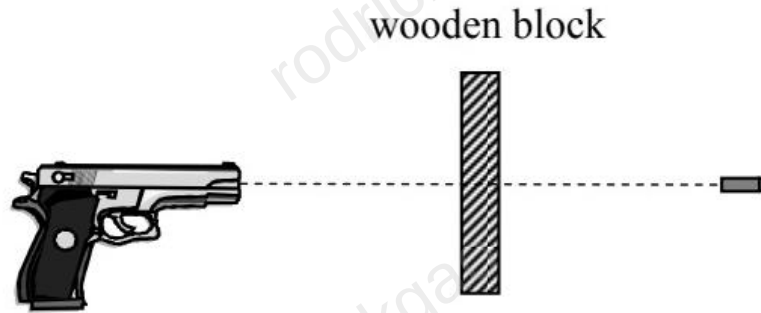
D. (2) and (3) only

Official answer: B

A direction reversal requires velocity to change sign. At 1 s, P still has negative velocity; only its acceleration changes sign. P reverses at 2 s, so (1) is false.

Signed areas give $x_P(2) = -\frac{1}{2}(2)(2) = -2$ m and $x_Q(2) = \frac{1}{2}(2)(2) = 2$ m: separation 4 m, so (2) is true. At 4 s, $x_P = -2 + \frac{1}{2}(2)(4) = 2$ m, while $x_Q = \frac{1}{2}(4)(4) = 8$ m. Equal velocities at 4 s do not mean equal positions. Only (2): B.

Question 12



A 50 g bullet fired at 400 m s^{-1} passes through a fixed wooden block of thickness 6 cm and emerges at 250 m s^{-1} . Find the average resistive force of the block on the bullet. Neglect air resistance and gravity.

A. $4.06 \times 10^4 \text{ N}$

B. $1.02 \times 10^4 \text{ N}$

C. 125 N

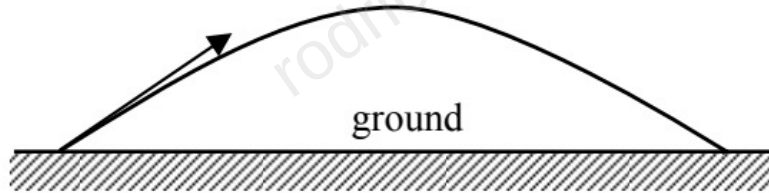
D. Cannot be found because the travel time inside the block is unknown.

Official answer: A

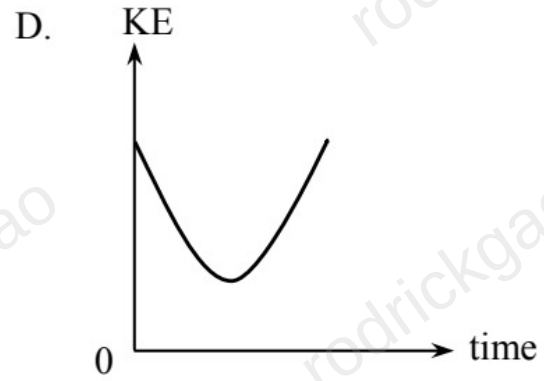
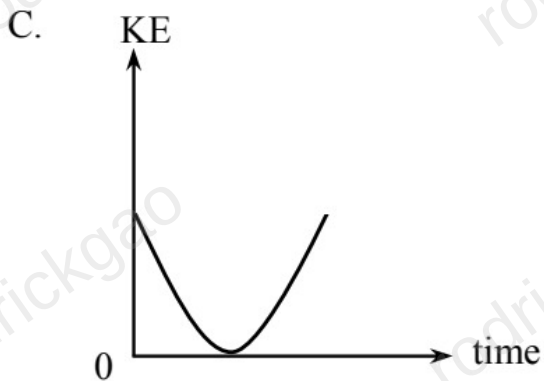
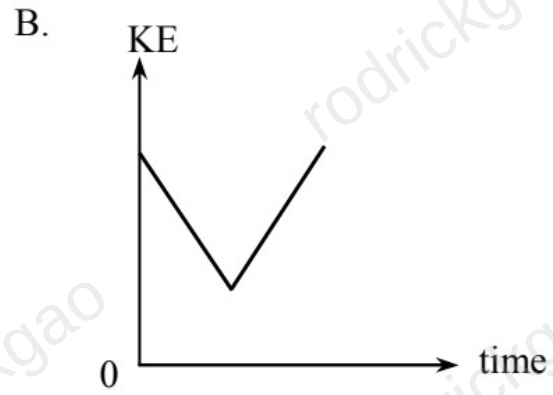
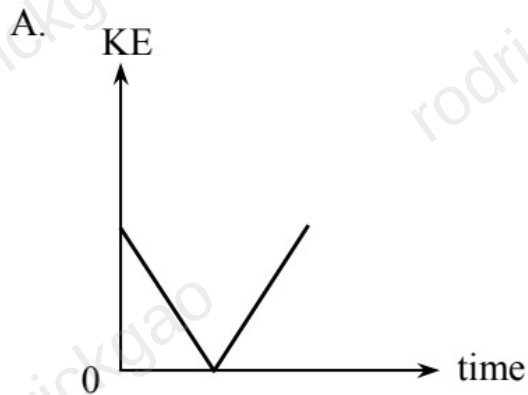
Convert $m = 0.050 \text{ kg}$ and $d = 0.060 \text{ m}$. Work against resistance equals the loss of kinetic energy: $F_{\text{avg}}d = \frac{1}{2}m(u^2 - v^2)$.

$F_{\text{avg}} = 0.5(0.050)(400^2 - 250^2)/0.060 = 40625 \text{ N} \approx 4.06 \times 10^4 \text{ N}$, opposite to motion: A. Time is unnecessary for this work-energy calculation.

Question 13



* A particle is projected at $t = 0$ and follows the parabolic path shown before landing. Which graph represents its kinetic energy (KE) against time before landing? Neglect air resistance.



Official answer: D

Horizontal velocity u_x is constant; vertical velocity is $u_y - gt$. Thus $KE(t) = \frac{1}{2}m[u_x^2 + (u_y - gt)^2]$, a smooth upward-opening parabola.

At the highest point the vertical velocity is zero but the horizontal velocity is not, so the minimum KE is positive. The curve is D, not the V-shaped B or the zero-minimum C.

Question 14



A semicircular cardboard hangs at O from a spring balance reading 5 N. Which statements are correct?

- (1) Its weight is 5 N.
- (2) Its centre of gravity is directly under O.
- (3) The reading would become zero on the Moon's surface.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

In static equilibrium the upward balance force equals the weight: 5 N. Zero torque about the suspension point requires the weight line to pass vertically through O, with the centre of gravity below O in this stable hanging position.

The Moon has a nonzero gravitational field, so the reading decreases but is not zero. Statements (1) and (2) only:
A.

Question 15

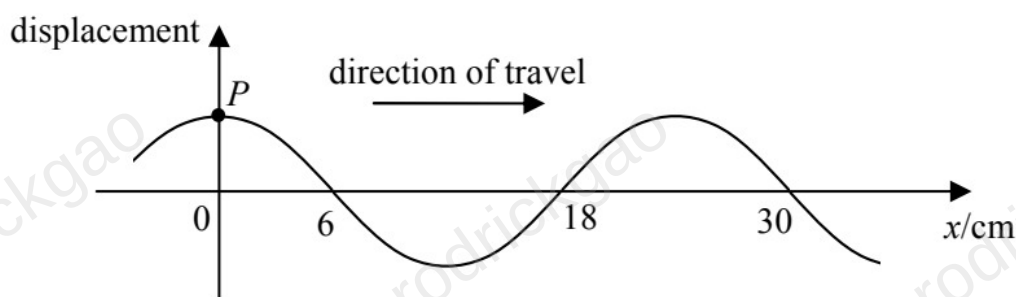
* The mass of Mars is about $1/10$ of Earth's mass and its radius about $1/2$ of Earth's radius. In terms of Earth's surface gravitational acceleration g , the approximate surface gravitational acceleration on Mars is

- A. $0.2g$
- B. $0.4g$
- C. $2.5g$
- D. $4g$

Official answer: B

Surface gravitational acceleration is GM/R^2 . The ratio is $g_{\text{Mars}}/g = (1/10)/(1/2)^2 = 0.4$, so B.

The radius ratio must be squared; using only inverse radius would incorrectly give $0.2g$.

Question 16

The snapshot shows a continuous transverse wave travelling along x at $t = 0$. At $t = 1.5$ s, particle P passes its equilibrium position for the second time. Find the wave speed.

- A. 20 cm s^{-1}
- B. 12 cm s^{-1}
- C. 6 cm s^{-1}
- D. 4 cm s^{-1}

Official answer: B

The equivalent downward zero crossings are at $x = 6$ and 30 cm, giving $\lambda = 24$ cm. P begins at a crest and passes equilibrium at $T/4$ and $3T/4$.

Thus $3T/4 = 1.5$ s, $T = 2.0$ s and $v = \lambda/T = 12 \text{ cm s}^{-1}$: B. The second crossing is not at $T/2$.

Question 17

7.

Figure (1)

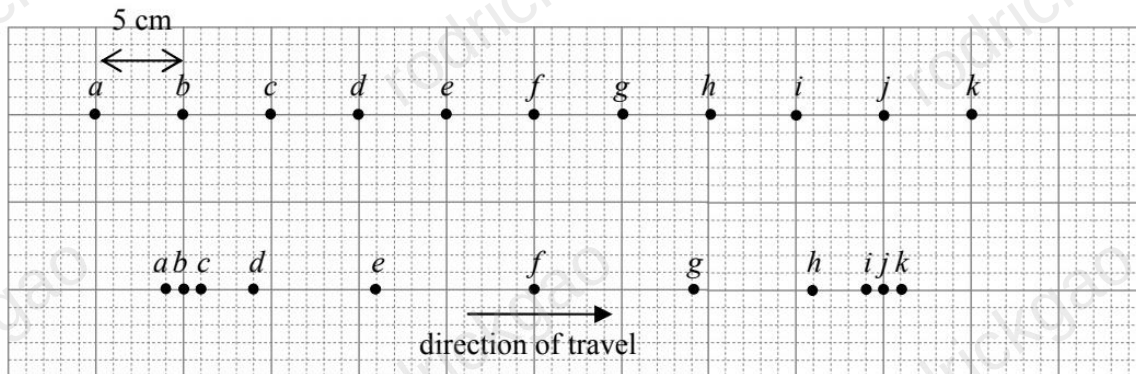


Figure (1) shows equilibrium positions a to k, separated by 5 cm. A longitudinal wave travels right at 80 cm s^{-1} . Figure (2) shows the particle positions at one instant. Find amplitude and frequency.

A. 6 cm; 2 Hz

B. 6 cm; 4 Hz

C. 9 cm; 2 Hz

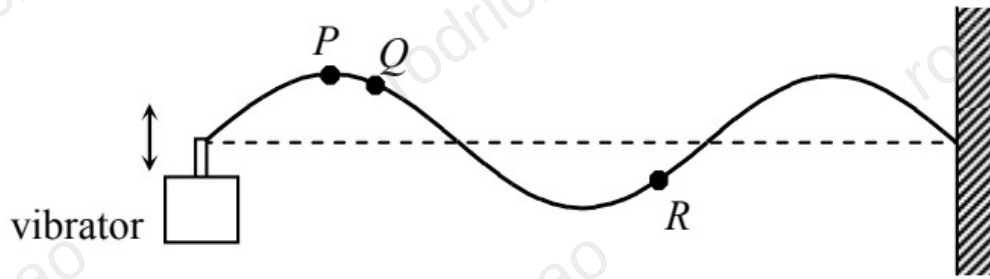
D. 9 cm; 4 Hz

Official answer: A

Compare each labelled particle with its own equilibrium position. The largest displacement is 6 cm (e.g. d or h), not the separation between neighbouring particles.

Consecutive compression centres are b and j, eight equilibrium spacings apart: $\lambda = 8(5) = 40 \text{ cm}$. Therefore $f = v/\lambda = 80/40 = 2 \text{ Hz}$; amplitude 6 cm and frequency 2 Hz: A.

Question 18



A vibrator generates a stationary wave on a string fixed at one end. The figure shows the string at one instant. Which descriptions must be correct?

- (1) P and Q are momentarily at rest at this instant.
- (2) Q and R take the same time to reach their respective equilibrium positions.
- (3) P and R are always in antiphase.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

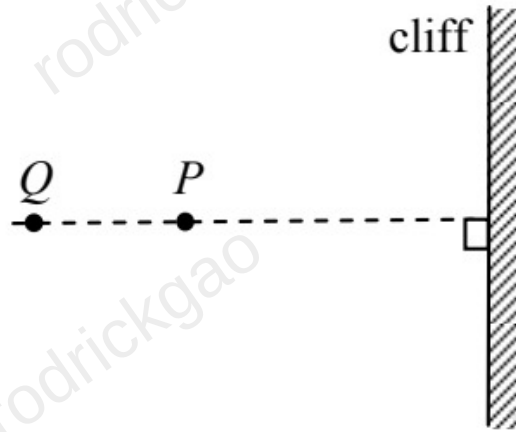
Official answer: D

A stationary-wave snapshot does not establish that the oscillation is at its maximum displacement. P is at a spatial crest, but that alone does not establish zero instantaneous particle velocity; (1) need not hold.

All non-node particles pass equilibrium simultaneously. Particles in adjacent loops, such as P and R, oscillate in antiphase. Thus (2) and (3) must hold: D. Different amplitudes do not imply different periods.

Question 19

Top view



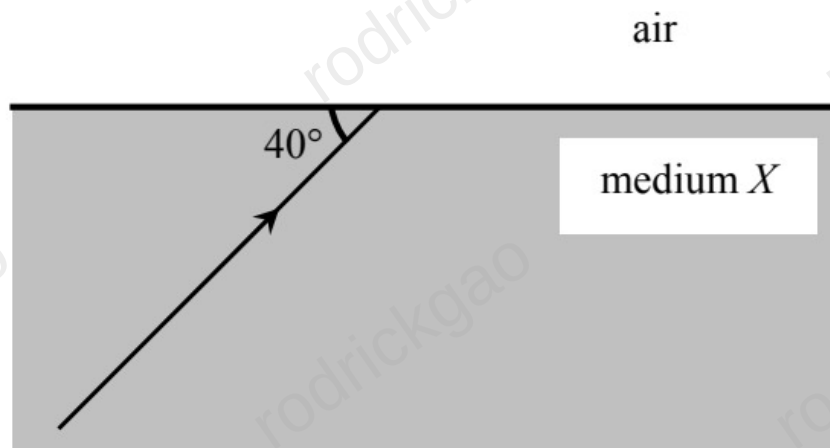
On a planet, astronauts P and Q stand 400 m and 600 m from a vertical cliff respectively, as shown in top view. P claps once and Q hears two sounds 4 s apart. Find the speed of sound in the planet's atmosphere.

A. 100 m s^{-1} B. 150 m s^{-1} C. 200 m s^{-1} D. 250 m s^{-1} **Official answer: C**

The direct sound travels $PQ = 600 - 400 = 200 \text{ m}$. The reflected sound travels P to cliff to Q: $400 + 600 = 1000 \text{ m}$.

The 4 s delay corresponds to the extra $1000 - 200 = 800 \text{ m}$, so $v = 800/4 = 200 \text{ m s}^{-1}$: C. It is neither the full echo travel time nor the travel time from the cliff alone.

Question 20



A ray travels from transparent medium X to air at 40° to the boundary plane. The angle between the refracted ray in air and reflected ray in X is 70° . Find the refractive index of X.

- A. $\sin 40^\circ / \sin 30^\circ$
- B. $\sin 30^\circ / \sin 40^\circ$
- C. $\sin 60^\circ / \sin 50^\circ$
- D. $\sin 50^\circ / \sin 60^\circ$

Official answer: C

Angles in Snell's law are measured from the normal: $i = 90^\circ - 40^\circ = 50^\circ$. The reflected ray is 40° below the interface, so the refracted ray is $70^\circ - 40^\circ = 30^\circ$ above it, giving $r = 60^\circ$.

$n_X \sin 50^\circ = 1 \sin 60^\circ$; hence $n_X = \sin 60^\circ / \sin 50^\circ$: C. The result is greater than one, as expected for the stated medium-to-air refraction.

Question 21

White light is resolved into component colours by a glass prism. Which statements are correct?

- (1) The refractive indices of glass for different component colours are not the same.
- (2) Red light travels faster than violet light in a vacuum.
- (3) All component frequencies decrease on entering the prism.

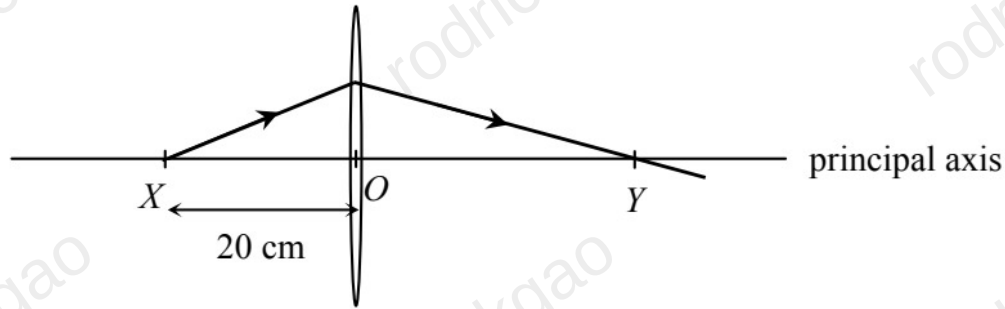
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

Dispersion occurs because glass has different refractive indices for different frequencies, producing different deviations: (1) is true.

All colours travel at c in a vacuum, so (2) is false. Crossing a stationary boundary changes speed and wavelength, not frequency, so (3) is false. Only (1): A.

Question 22



A point source X on the principal axis of a thin convex lens emits a ray that reaches the axis at Y after refraction. O is the optical centre; $OX = 20$ cm and $OY > OX$. Which statements are correct?

- (1) The focal length is shorter than 20 cm.
- (2) Moving the source away from the lens increases OY .
- (3) An object at Y gives a diminished image at X .

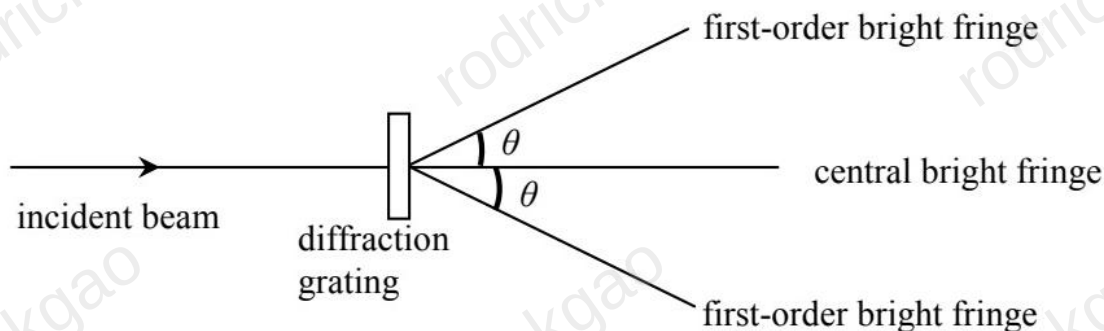
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

A real image forms with $u = 20$ cm and $v > 20$ cm. From $1/f = 1/u + 1/v$, $f < u$, so (1) is true. Increasing u increases $1/v$ and decreases v , so (2) is false.

By reversibility, a source at Y images at X . The magnification is $OX/OY < 1$, hence diminished. Statements (1) and (3): C.

Question 23



* Monochromatic light forms bright fringes after passing through a diffraction grating. Which arrangement gives the greatest angle θ between the central and first-order bright fringes? Options give grating lines per mm and colour.

A. 400; green

B. 400; blue

C. 200; green

D. 200; blue

Official answer: A

For first order, $d \sin \theta = \lambda$, so $\sin \theta = \lambda/d$. More lines per mm means a smaller spacing d , increasing θ .

Green light has a longer wavelength than blue light. Combine the longer wavelength with 400 lines per mm: A.

Do not confuse line density with slit spacing.

Question 24

Two small identical metal spheres X and Y carry $-2Q$ and $+6Q$. At their separation the electrostatic force magnitude is F .



They touch and are returned to their original positions. The force becomes

A. $F/4$, attractive

B. $F/4$, repulsive

C. $F/3$, attractive

D. $F/3$, repulsive

Official answer: D

Total charge is $-2Q + 6Q = 4Q$. Identical spheres share it equally, leaving $+2Q$ on each, so the new force is repulsive.

At the same separation, $F'/F = |(2Q)(2Q)|/|(-2Q)(6Q)| = 4/12 = 1/3$. Thus $F/3$, repulsive: D.

Question 25

* Lightning may occur when the uniform electric field between a thundercloud and ground reaches $3 \times 10^6 \text{ N C}^{-1}$. A flash discharges about 20 C. The cloud is 500 m high. Estimate the order of magnitude of the energy released.

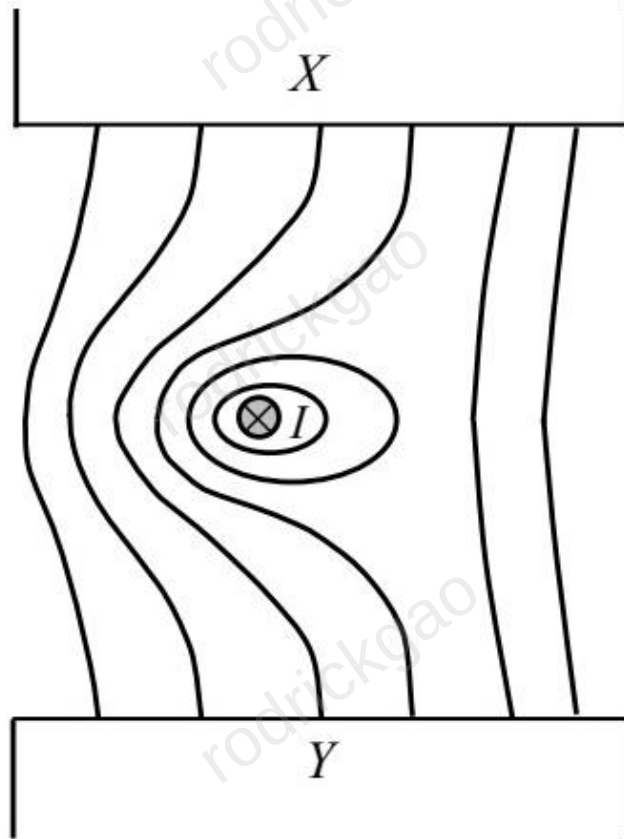
- A. 10^6 J
- B. 10^8 J
- C. 10^{10} J
- D. 10^{12} J

Official answer: C

Uniform-field potential difference $V = Ed = (3 \times 10^6)(500) = 1.5 \times 10^9 \text{ V}$.

Energy scale $QV = 20(1.5 \times 10^9) = 3 \times 10^{10} \text{ J}$, giving the listed order 10^{10} J : C. This is an order estimate, not a detailed time-dependent discharge model.

Question 26



A straight wire carrying current into the paper lies between magnetic pole pieces X and Y. The resultant field is shown. Find the polarity of X and direction of magnetic force on the wire. Ignore Earth's field.

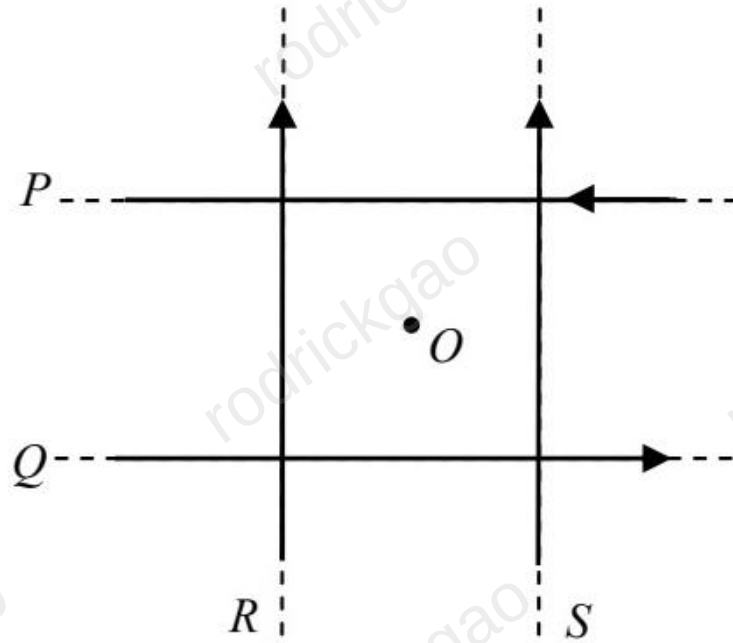
- A. N; right
- B. N; left
- C. S; right
- D. S; left

Official answer: C

The right-hand grip rule gives a clockwise wire field: upward to the left of the wire. The field lines are crowded on the left, so the external field reinforces this upward direction, from Y towards X. X is therefore S.

For current into the paper and external field upward, Fleming's left-hand rule ($I\ell \times B$) gives force to the right: C. Use the external field for the wire force, not its own circular field.

Question 27



Four insulated long straight wires P, Q, R and S in one plane carry equal currents as shown. O is in the plane and equidistant from each wire. Removing which wire increases the magnetic field strength at O?

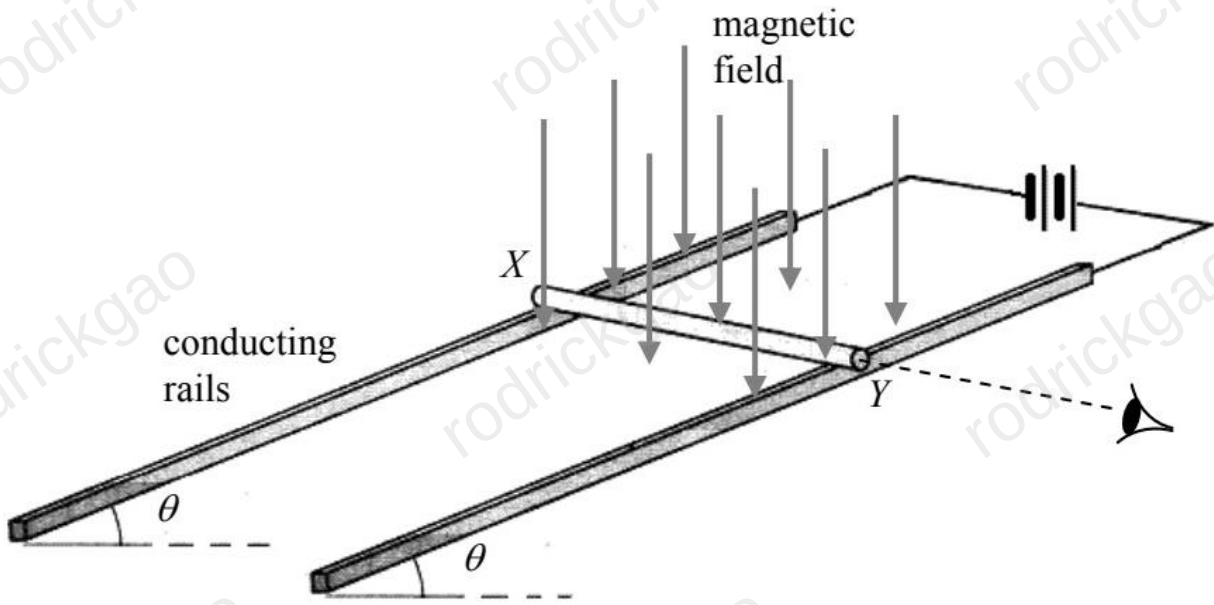
- A. P
- B. Q
- C. R
- D. S

Official answer: C

Equal current and distance give equal field magnitudes B_0 . Apply the right-hand grip rule at O: P, Q and S produce fields out of the paper, while R produces a field into the paper.

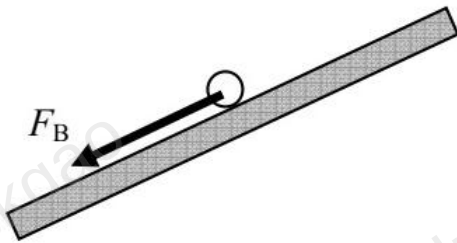
Initially the resultant is $3B_0 - B_0 = 2B_0$ outward. Removing R increases it to $3B_0$; removing any other wire decreases it. Choose C.

Question 28

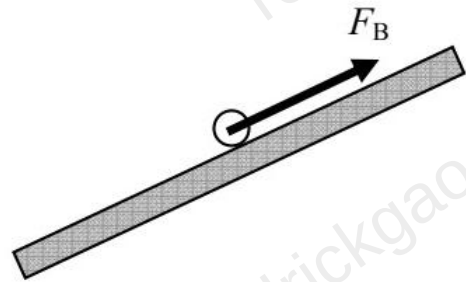


A copper rod XY lies on smooth conducting rails inclined at θ to the horizontal. A magnetic field acts vertically down and a battery is connected as shown. Which diagram gives the magnetic force F_B viewed from end Y ?

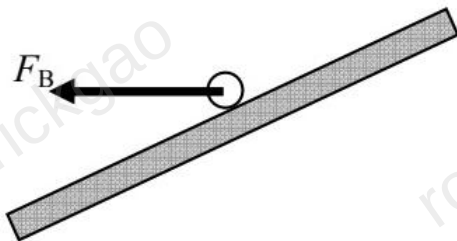
A.



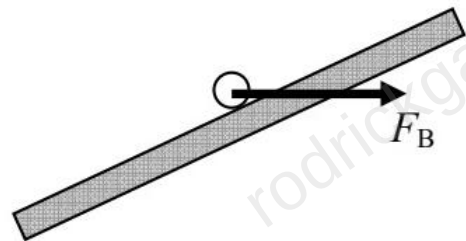
B.



C.



D.

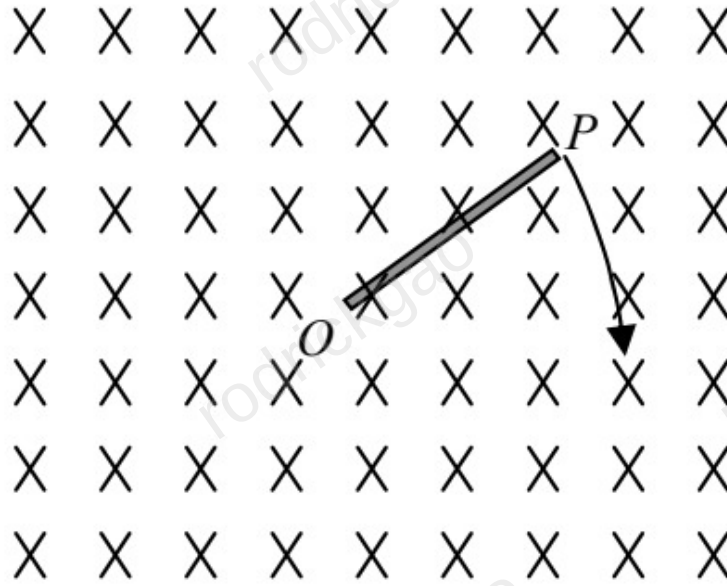


Official answer: C

The battery's positive (long-line) terminal is connected to the Y rail, so conventional current passes from Y to X . Viewed from Y towards X , current goes into the view.

With B vertically down, $I\ell \times B$ is horizontal to the left: C. The magnetic force must be perpendicular to B ; it is not along the sloping rails. Components of force, rather than the entire magnetic force, act along the rails.

Question 29



A metal rod OP rotates clockwise about O in the paper plane in a uniform magnetic field into the paper. Which statement is correct?

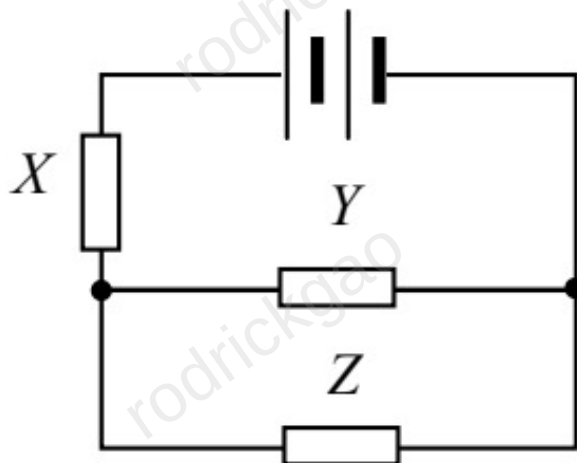
- A. Induced current flows from O to P .
- B. Induced current flows from P to O .
- C. An e.m.f. is induced with O at higher potential.
- D. An e.m.f. is induced with P at higher potential.

Official answer: D

For a positive charge moving clockwise with the rod, $qv \times B$ is radially outward, towards P . Electrons move towards O , leaving P at higher potential.

Charge separation establishes an induced e.m.f., but the isolated rod has no closed circuit for a sustained current. Thus D, not a current-flow option.

Question 30



Identical resistors X, Y and Z are connected as shown. A battery of negligible internal resistance supplies total power 24 W. Find the power dissipated in Z.

A. 3 W

B. 4 W

C. 6 W

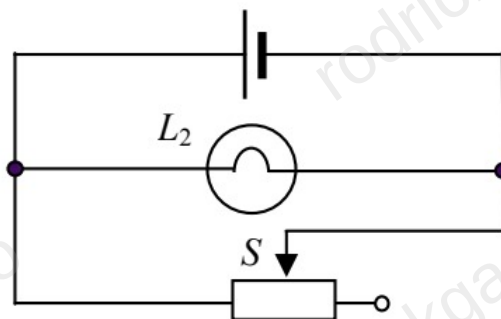
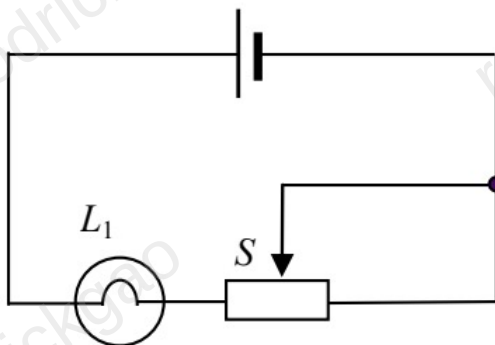
D. 8 W

Official answer: B

X is in series with the parallel pair Y and Z. If each branch current is i , X carries $2i$. Since $P = I^2R$, $P_X : P_Y : P_Z = 4 : 1 : 1$.

Z therefore dissipates one sixth of the total: $24/6 = 4$ W, B. Equal resistances do not imply equal powers when their currents differ.

Question 31



In both circuits the cells have constant e.m.f. and negligible internal resistance. As each rheostat slider S moves right from its mid-position, how do the brightnesses of L_1 and L_2 change?

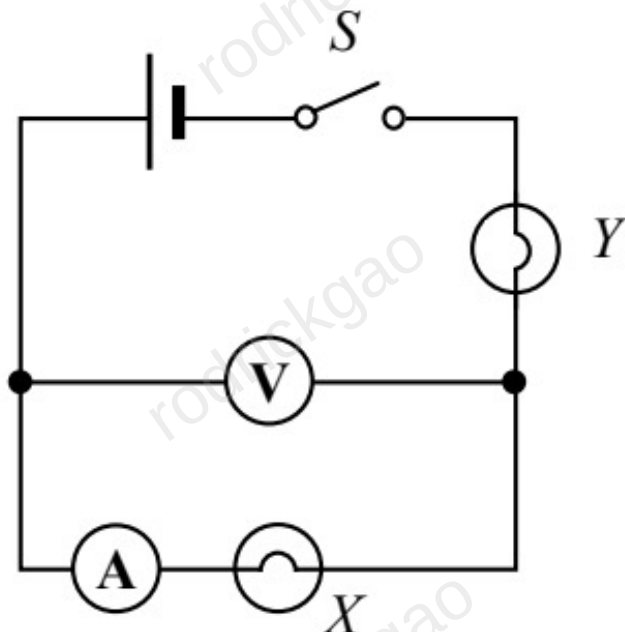
- A. L_1 dimmer; L_2 unchanged
- B. L_1 dimmer; L_2 brighter
- C. L_1 unchanged; L_2 dimmer
- D. L_1 brighter; L_2 unchanged

Official answer: A

In the left circuit the section to the right of the slider is bypassed. Moving S right lengthens the effective series resistance, reducing current and making L_1 dimmer.

L_2 is directly across an ideal cell; changing the parallel rheostat changes total current but not the bulb voltage. L_2 remains unchanged: A.

Question 32



The cell has negligible internal resistance. Closing S leaves both bulbs unlit; the voltmeter has a reading but the ammeter reads zero. If there is only one fault, which is possible?

- A. Bulb X is short-circuited.
- B. Bulb Y is short-circuited.
- C. Bulb X is burnt out and open-circuit.
- D. Bulb Y is burnt out and open-circuit.

Official answer: C

If X is open, the ammeter branch is broken, so it reads zero. With an ideal high-resistance voltmeter, negligible current passes through Y, so Y is unlit and has negligible voltage drop; the voltmeter measures approximately the cell e.m.f.

If Y were open, the source would be disconnected and the voltmeter would read zero. Shorting either bulb would allow current through the other bulb. The consistent fault is X open: C.

Question 33

Which domestic appliance consumes a power close to 1 kW under normal working conditions?

- A. An electric fan
- B. A microwave oven
- C. A fluorescent lamp
- D. A TV set

Official answer: B

1 kW = 1000 W. In this question's typical domestic-appliance comparison, the microwave oven is the kilowatt-scale heating appliance: B.

The fan, fluorescent lamp and TV in the intended comparison have substantially smaller power demands. This is a scale comparison, not a universal rating for every model.

Question 34

${}^{238}_{92}\text{U}$ undergoes α - β - β - α decay to become nuclide X. Find the atomic number and mass number of X, in that order.

A. 90; 230

B. 90; 234

C. 88; 230

D. 88; 234

Official answer: A

Each α decay reduces mass number A by 4 and atomic number Z by 2. Each β^- decay leaves A unchanged and increases Z by 1.

Two α and two β^- decays give $A = 238 - 8 = 230$ and $Z = 92 - 4 + 2 = 90$. In the requested order Z, A the answer is 90, 230: A.

Question 35

* Polonium-210 is a pure α -emitter with half-life 140 days and decays to stable lead. A sample initially contains 420 mg of pure polonium-210. Estimate the mass of polonium-210 remaining after 70 days.

A. 315 mg

B. 297 mg

C. 210 mg

D. 105 mg

Official answer: B

The remaining isotope mass follows $m = m_0 2^{-t/T}$. Thus $m = 420 \times 2^{-70/140} = 420/\sqrt{2} \approx 297$ mg: B.

Radioactive decay is exponential, not linear. Half of a half-life does not mean half of the usual 50% loss, so 315 mg is incorrect. The question asks for remaining polonium, not total polonium plus lead.

Question 36

* The Sun releases energy by thermonuclear fusion while its mass decreases. Its average power is 3.8×10^{26} W. Estimate its mass decrease in one second.

A. 4.2×10^6 kg

B. 4.2×10^9 kg

C. 1.3×10^{15} kg

D. 1.3×10^{18} kg

Official answer: B

In 1 s the energy released is $E = Pt = 3.8 \times 10^{26}$ J. Mass-energy equivalence gives $\Delta m = E/c^2$.

Using $c = 3.0 \times 10^8 \text{ m s}^{-1}$, $\Delta m = (3.8 \times 10^{26})/(9.0 \times 10^{16}) = 4.22 \times 10^9$ kg: B. Square the entire speed of light, including its power of ten.